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Getting started with R – Frequently Asked Questions

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1 Installation

1.1 Can I install R on my home computer or laptop?

Yes. R is open-source software. It is available for Windows, Mac, and Linux.

For Windows:

1. Go to <https://cran.r-project.org/>.
2. Click on 'Download R for Windows'.
3. Click on 'base'.
4. Click on 'Download R for Windows'. This will download an .exe file.
5. Double-click on the downloaded .exe file. This will install R on your computer. Installation is straightforward, you can accept all the defaults.

There should now be an 'R 3...' icon on your desktop. Double-click and R opens.

2 Alternative editors – RStudio

It is possible to do everything just using the plain R editor and user interface. However, RStudio provides some extras (e.g. writing reports and documents) and is very popular. Note that the statistical software is still R, RStudio just makes it easier to read and write the code and easier to interact with R.

RStudio is open-source and can be found at <https://www.rstudio.com/>.

Note that R should be installed first.

3 Where do I start to learn R?

The only way to learn R is to use it. Here are some tutorials:

1. <https://www.rstudio.com/online-learning/R>
2. In the R console type `help.start()`. 'An Introduction to R' is an excellent place to start. Go to Appendix A: A sample session.
3. *An R Companion to Introstat*. Available on Vula (Resources > R Resources). Shows you how to do all the stuff from first year using R.
4. <http://data.princeton.edu/R/default.html>
5. Online introductory course on lynda: <http://lynda.uct.ac.za>: *Up and Running with R*.
6. Start with using the code given to you in class, copy it and adapt it to your problem.

3.1 Online tutorials and documentation

1. Go to <http://cran.r-project.org/>. On the left under Documentation, click on 'Contributed'. Especially useful may be: *Practical Regression and Anova using R*, by Julian Faraway.
2. Data Camp: <https://www.datacamp.com/>
3. Quick-R: <http://www.statmethods.net/index.html>.
4. R Cookbook: <http://www.cookbook-r.com/>
5. <https://www.programiz.com/>
6. The R help can be useful if you know the name of the R function and need more specifics on what to input as arguments. For example, `help(lm)` will give you all the details on how the `lm` (for linear model) function in R should be called.

3.2 Books available in UCT library

- Braun, J. & Murdoch, D. J. (2007). A first course in statistical programming with R. Cambridge University Press.
- Crawley, M. J. 2009. The R book. Wiley.
- Dalgaard, P. (2002). Introductory Statistics with R. Springer.
- Fox, J. (2002). An R and S-PLUS companion to Applied Regression. Sage.
- Verzani, J. (2005). Using R for Introductory Statistics. Chapman & Hall. (519.50285 VERZ)
- Venables, W. N. & Ripley, B. D. (2002). Modern Applied Statistics with S. New York: Springer, 4th ed. (*more advanced*)
- Zuur, A. F, Ieno, E. N. and Meesters, E. H. W. G. (2009). A Beginner's Guide to R. Springer. (519.50285 ZUUR)

4 Saving my work

Being able to save code is an enormous advantage.

- You don't have to start from scratch each time.
- You can copy and change code for new projects!
- Next time you need to do the same analysis with a different data set, just rerun the code!
- You have a record of exactly what you did!

4.1 What should I be saving?

You only need to save the *script* file! These should be saved with an .R extension for easy recognition.

Note that you don't need to save the R output from the console.

If you need some part of the output for a report, rerun the code and copy that part.

4.2 How should I save plots and graphics?

For inclusion in MS Word documents: Once you have created the graph, in RStudio, click on 'Export' > 'Save as Image'. Use the **SVG image format**! Everything else looks terrible in Word.

For LaTeX, .pdf and .ps work well.

5 R packages

There are many (thousands) add-on packages to do special things for R.

5.1 How do I load an R package?

If the package is already installed on your computer just load it with `library(package)`, where you should replace package with the package's name.

In the UCT Scilabs most packages that you should need are already installed. If not see below.

5.2 Installing a new R package?

```
install.packages(package)
```

At UCT you will have to do this every time you log into a computer because the computers are 'cleaned' overnight.

Replace package with your package name, in inverted commas, e.g. `install.packages("vegan")`. At home you will only have to do this once.

6 Where are my files?

I don't know, but for the future, the first thing you should do whenever you start a new R session is to **change your working directory** to where you *want* your files to be.

In the basic R editor go to the File menu: File > change working directory (e.g. a folder on your flash drive called STATS). Keep all your programs, data files in this directory. R will then also save

all other files, such as graphics to this directory. Give meaningful names to your files to make it easy to guess their content.

In RStudio navigate to your working directory (under the Files tab on the RHS). Then under the 'More' option (with a little cog wheel) click on 'Set as Working directory'.

You can also then copy this command (echoed in the console) into your script file for future use.

7 How do I get my data file into R?

The most common file types you will encounter are .xlsx (or .xls), .txt and .csv. Note that the command to use depends on the file type.

1. **.csv files:** `mydata <- read.csv("mydata.csv")`
2. **.xls files:** In Excel save the file as a .csv file (comma separated).
Then in R: `mydata <- read.csv("mydata.csv")`
3. **.xls files:** In Excel save the file as a .csv file (comma separated).
Then in R: `mydata <- read.csv("mydata.csv")`
4. **.xls files:** In Excel save the file as a .csv file (comma separated).
Then in R: `mydata <- read.csv("data.csv", header = TRUE)`
5. **.xlsx files:** Suppose your data are in a file called `mydata.xlsx`.

```
library(xlsx)
mydata <- read.xlsx("mydata.xlsx", sheetName = "Sheet1")
```

6. **.txt files:** In R: `worms <- read.txt("worms.txt", header = TRUE)`

A few things to note:

- Make sure the file is in your working directory and that you have told R what your working directory is.
- The `header = TRUE` part is for when the first row in your data file contains headings or variable names. If not, set this to `FALSE`.
- File names must always go into inverted commas.
- If you don't read it into something (e.g. into 'mydata' here), R does not keep it in memory, i.e. you cannot refer to it or work on it.